# HW3

Monday, January 27, 2020 1:05 PM

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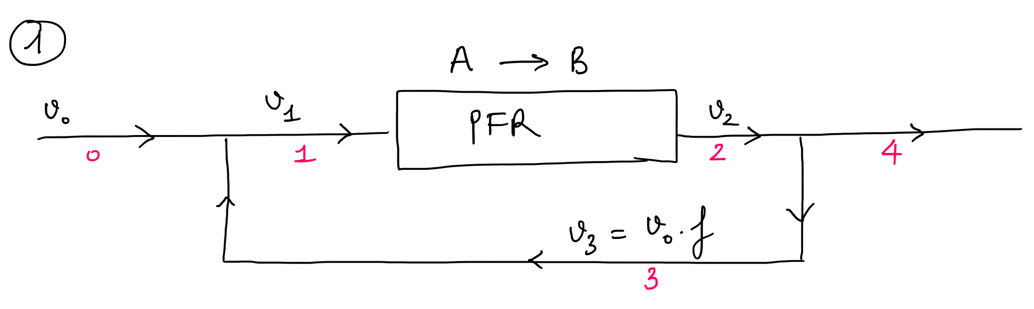


- $X_T = \frac{C_{A_0} - C_{A_4}}{C_{A_0}} = 1 - \frac{C_{A_4}}{C_{A_0}}$
- $C_{A_4} = C_{A_2} = C_{A_1}(1-X)$ → find relationship between $C_{A_1}$ & $C_{A_0}$ (1)
  find X (2)

(1) $F_{A_0} + F_{A_3} = F_{A_1}$

$\Rightarrow C_{A_0} v_0 + C_{A_3} v_3 = C_{A_1} v_1$

$\Rightarrow C_{A_0} v_0 + C_{A_2} v_0 f = C_{A_1} v_0 (1+f)$ $[C_{A_3} = C_{A_2}]$

$\Rightarrow C_{A_0} + C_{A_2} f = C_{A_1}(1+f)$

$\Rightarrow C_{A_0} + C_{A_1}(1-X) f = C_{A_1}(1+f)$ $[C_{A_2} = C_{A_1}(1-X)]$

$\Rightarrow C_{A_1}[(1+f) - (1-X)f] = C_{A_0}$

$\Rightarrow C_{A_1}(Xf + 1) = C_{A_0}$

$\Rightarrow \boxed{C_{A_1} = \frac{C_{A_0}}{1+Xf}}$

(2) PFR: $\frac{dF_A}{dV} = r_A$

$F_A = F_{A_1}(1-X) = C_{A_1} v_1 (1-X) = C_{A_1} v_0 (f+1)(1-X) = \frac{C_{A_0}}{1+Xf} v_0 (f+1)(1-X)$

$r_A = -k C_A = -k C_{A_1}(1-X) = -k \frac{C_{A_0}}{1+Xf}(1-X)$

$\Rightarrow \frac{d}{dV}\left[\frac{C_{A_0}}{1+Xf} v_0 (f+1)(1-X)\right] = -k \frac{C_{A_0}}{1+Xf}(1-X)$

$\Rightarrow \cancel{C_{A_0}} \cdot v_0 (1+f) \frac{d}{dV}\left[\frac{1-X}{1+Xf}\right] = -k \cancel{C_{A_0}} \frac{(1-X)}{1+Xf}$

$\Rightarrow \frac{d}{dV}\left[\frac{1-X}{1+Xf}\right] = \frac{-k(1-X)}{(1+Xf)\, v_0 (1+f)}$

$\Rightarrow \frac{-\frac{dX}{dV}(1+Xf) - f\frac{dX}{dV}(1-X)}{(1+Xf)^2} = \frac{-k(1-X)}{(1+Xf)\, v_0 (1+f)}$

$\Rightarrow \frac{dX}{dV}\left[\frac{(1+Xf) + f(1-X)}{(1+Xf)^{\cancel{2}}}\right] = \frac{k(1-X)}{\cancel{(1+Xf)}\, v_0 (1+f)}$

$\Rightarrow dX\left[\frac{1+f}{(1+Xf)(1-X)}\right] = \frac{k}{v_0(1+f)} dV$

$\Rightarrow \int_0^X \frac{dX}{(1+Xf)(1-X)} = \frac{k}{v_0(1+f)^2} \int_0^V dV$

- $\frac{f}{1+Xf} + \frac{1}{1-X} = \frac{f - fX + 1 + Xf}{(1+Xf)(1-X)} = \frac{f+1}{(1+Xf)(1-X)}$

$\Rightarrow \frac{1}{1+f} \int_0^X \left(\frac{f}{1+Xf} + \frac{1}{1-X}\right) dX = \frac{k}{v_0(1+f)^2} V$

$\Rightarrow \ln(1+Xf) - \ln(1-X) \Big|_0^X = \frac{kV}{v_0(1+f)}$

$\Rightarrow \ln\frac{(1+Xf)}{(1-X)} = \frac{kV}{v_0(1+f)}$

$\Rightarrow \frac{1+Xf}{1-X} = \underbrace{\exp\left(\frac{kV}{v_0(1+f)}\right)}_{A}$

$\Rightarrow 1 + Xf = A(1-X) = A - AX$

$\Rightarrow X(f+A) = A - 1$

$\Rightarrow X = \frac{A-1}{f+A}$

(1) (2) $\Rightarrow X_T = 1 - \frac{C_{A_1}(1-X)}{C_{A_0}} = 1 - \frac{C_{A_0}}{(1+Xf)} \frac{(1-X)}{C_{A_0}}$

$= 1 - \frac{1-X}{1+Xf}$

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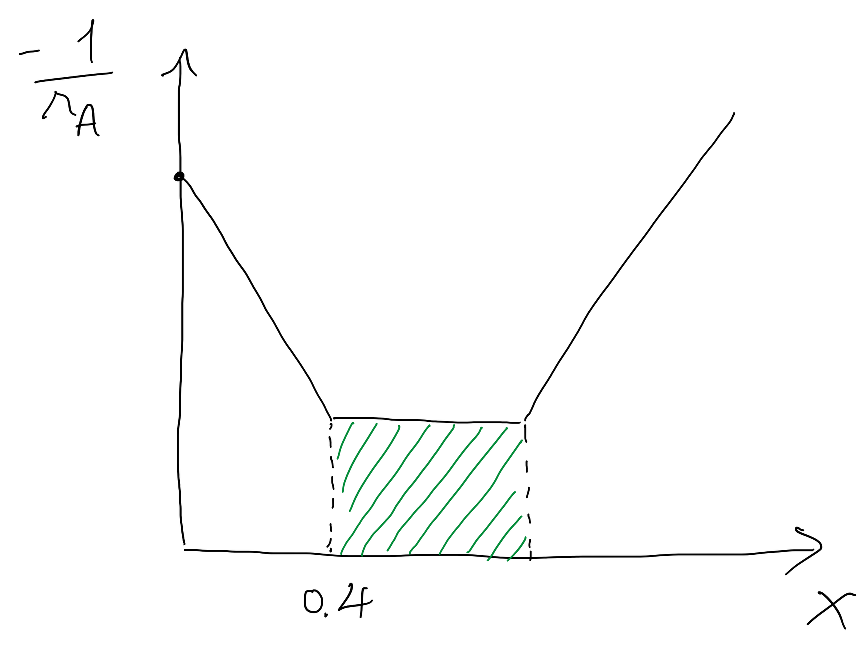
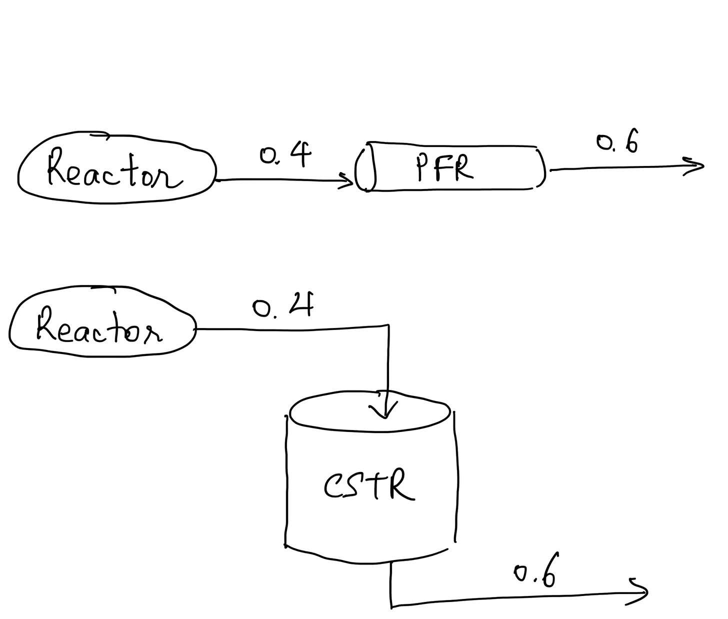
| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| X | 0 | 0.2 | 0.4 | 0.45 | 0.5 | 0.6 | 0.8 | 0.9 |
| r | 1 | 1.67 | 5 | 5 | 5 | 5 | 1.25 | 0.91 |
| 1/-r | 1 | 0.6 | 0.2 | 0.2 | 0.2 | 0.2 | 0.8 | 1.1 |




(a) $V_{CSTR} = \frac{F_{A_0} X}{-r_A} = \frac{300 \times 0.4}{5} = 24\ dm^3$

$V_{PFR} = F_{A_0} \int_0^X \frac{dX}{-r_A} = 72\ dm^3$. (area under the curve)

(b)




Because the rate is constant over this conversion range.